COHOMOLOGY OF A QUASI-COHERENT SHEAF IS INDEPENDENT OF THE TOPOLOGY USED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries the author's own slogan; the chapter's preamble defines slogan environments as comment environments, so the slogan is part of the counted source text and is not typeset.

Editorial note. The proof contains the author's sentence suggesting that the reader may skip the rest of the proof or rewrite it for a clearer exposition; the sentence is reproduced verbatim, and the derivation that follows is complete as written, so it is a reading suggestion rather than an omission.

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Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Vanishing criterion for cohomology on a basis of objects with a cofinal system of coverings. Cohomology on Sites, Lemma sites-cohomology-lemma-cech-vanish-collection, invoked at proof lines 3018-3024. Supplies the vanishing of higher cohomology of the associated module on affines.
- Standard affine coverings compute the Cech cohomology of a quasi-coherent module. Same chapter, Lemma lemma-standard-covering-Cech-quasi-coherent, invoked at proof lines 3024-3026. Supplies assumption (3) of the vanishing criterion.
- Cech spectral sequence and its application to a covering. Cohomology on Sites, Lemma sites-cohomology-lemma-cech-spectral-sequence-application, invoked at proof lines 3039-3046. Supplies the comparison of Cech and ordinary cohomology on separated objects.
- Lifting a map of complexes out of an exact complex of injectives. Derived Categories, Lemma derived-lemma-morphisms-lift, invoked at proof lines 3066-3069. Supplies the comparison map between the two derived global section complexes.

Source reference. Original lemma 01XD: sources/stacks-chow/tags.txt, source lines 2514–2514. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 013P: sources/stacks-chow/tags.txt, source lines 1444–1444. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03FJ: sources/stacks-descent/pinned-original-source.tex, source lines 2948–2948. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03F7: sources/stacks-chow/tags.txt, source lines 4482–4482. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03F9: sources/stacks-chow/tags.txt, source lines 4484–4484. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Proposition 0.1. *Let S be a scheme. Let $\mathcal{F}$ be a quasi-coherent sheaf on S. Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$.*

(1) *There is a canonical isomorphism*
$$H^q(S, \mathcal{F}) = H^q((Sch/S)_\tau, \mathcal{F}^a).$$
(2) *There are canonical isomorphisms*
$$H^q(S, \mathcal{F}) = H^q(S_{Zar}, \mathcal{F}^a) = H^q(S_{étale}, \mathcal{F}^a).$$

Proof. The result for $q = 0$ is clear from the definition of $\mathcal{F}^a$. Let $\mathcal{C} = (Sch/S)_\tau$, or $\mathcal{C} = S_{étale}$, or $\mathcal{C} = S_{Zar}$.

We are going to apply Cohomology on Sites, Lemma 03F9 with $\mathcal{F} = \mathcal{F}^a$, $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ the set of affine schemes in $\mathcal{C}$, and $\mathrm{Cov} \subset \mathrm{Cov}_\mathcal{C}$ the set of standard affine τ-coverings. Assumption (3) of the lemma is satisfied by Lemma 03FJ. Hence we conclude that $H^p(U, \mathcal{F}^a) = 0$ for every affine object U of $\mathcal{C}$.

Next, let $U \in \mathrm{Ob}(\mathcal{C})$ be any separated object. Denote $f : U \to S$ the structure morphism. Let $U = \bigcup U_i$ be an affine open covering. We may also think of this as a τ-covering $\mathcal{U} = \{U_i \to U\}$ of U in $\mathcal{C}$. Note that $U_{i_0} \times_U \ldots \times_U U_{i_p} = U_{i_0} \cap \ldots \cap U_{i_p}$ is affine as we assumed U separated. By Cohomology on Sites, Lemma 03F7 and the result above we see that
$$H^p(U, \mathcal{F}^a) = \check{H}^p(\mathcal{U}, \mathcal{F}^a) = H^p(U, f^*\mathcal{F})$$
the last equality by Cohomology of Schemes, Lemma 01XD. In particular, if S is separated we can take $U = S$ and $f = \mathrm{id}_S$ and the proposition is proved. We suggest the reader skip the rest of the proof (or rewrite it to give a clearer exposition).

Choose an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$ on S. Choose an injective resolution $\mathcal{F}^a \to \mathcal{J}^\bullet$ on $\mathcal{C}$. Denote $\mathcal{J}^n|_S$ the restriction of $\mathcal{J}^n$ to opens of S; this is a sheaf on the topological space S as open coverings are τ-coverings. We get a complex
$$0 \to \mathcal{F} \to \mathcal{J}^0|_S \to \mathcal{J}^1|_S \to \ldots$$

which is exact since its sections over any affine open $U \subset S$ is exact (by the vanishing of $H^p(U, \mathcal{F}^a)$, $p > 0$ seen above). Hence by Derived Categories, Lemma 013P there exists map of complexes $\mathcal{J}^\bullet|_S \to \mathcal{I}^\bullet$ which in particular induces a map

$$R\Gamma(\mathcal{C}, \mathcal{F}^a) = \Gamma(S, \mathcal{J}^\bullet) \longrightarrow \Gamma(S, \mathcal{I}^\bullet) = R\Gamma(S, \mathcal{F}).$$

Taking cohomology gives the map $H^n(\mathcal{C}, \mathcal{F}^a) \to H^n(S, \mathcal{F})$ which we have to prove is an isomorphism. Let $\mathcal{U} : S = \bigcup U_i$ be an affine open covering which we may think of as a τ-covering also. By the above we get a map of double complexes

$$\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{J}) = \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{J}|_S) \longrightarrow \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{I}).$$

This map induces a map of spectral sequences

$${}^\tau E_2^{p,q} = \check{H}^p(\mathcal{U}, \underline{H}^q(\mathcal{F}^a)) \longrightarrow E_2^{p,q} = \check{H}^p(\mathcal{U}, \underline{H}^q(\mathcal{F}))$$

The first spectral sequence converges to $H^{p+q}(\mathcal{C}, \mathcal{F})$ and the second to $H^{p+q}(S, \mathcal{F})$. On the other hand, we have seen that the induced maps ${}^\tau E_2^{p,q} \to E_2^{p,q}$ are bijections (as all the intersections are separated being opens in affines). Whence also the maps $H^n(\mathcal{C}, \mathcal{F}^a) \to H^n(S, \mathcal{F})$ are isomorphisms, and we win. □